



HOSTELHUB

Student Portal Complete Feature & Architecture Master Blueprint

Student App 100% Complete Scope •
Hinglish

 **Document Overview:** Ye document HostelHub ke **Student Portal (frontend-user & Mobile App)** ke har ek function, feature, authentication security, real-time chat, complaint engine, mess ratings, aur mobile/desktop UI layout ka 100% complete master breakdown provide karta hai.

1 Student Authentication & Security Engine



Account Access & Onboarding Features

AUTH ENGINE

- **Email & Password Login:** Secure login with JWT Token storage in browser/app state for seamless sessions.
- **Google OAuth 2.0 Single Sign-On:** 1-Click "Sign in with Google" integration for instant login without typing passwords.
- **Student Registration Signup:** New student onboarding form requiring Full Name, Email, Password, Phone Number, Gender (Male/Female), and Batch selection. New accounts get `pending_verification` status until Admin/Warden approval.
- **Forgot Password & Email OTP Verification:** 6-Digit OTP sent to registered email for password recovery with countdown expiration timer.
- **In-App Change Password:** Secure password change from Profile settings page.

2 Student Profile & Account Details



Digital Student ID & Personal Details

PROFILE MANAGEMENT

- **Profile Overview Card:** View Name, Roll Number, Allocated Room Number, Hostel Block, Roll No, Batch (e.g. Batch 2025), Gender, Phone, and Bio.
- **Profile Picture Upload:** Upload and change profile avatar directly from mobile camera or gallery (Powered by backend Multer storage engine).
- **Editable Student Bio & Contact:** Personal bio and contact phone number updates.

3 Complaint Lodging & Real-Time Tracking Engine

Complaint Registration & Media Attachment CORE FEATURE

- **Multi-Category Complaint Registration:** Raise maintenance or misconduct issues under categories:
 -  **Plumbing:** Water leakage, tap repair, flush issue.
 -  **Electrical:** Fan, light bulb, switchboard, AC fault.
 -  **Wi-Fi / IT:** Network downtime, router issue, slow speed.
 -  **Mess & Food:** Food quality, hygiene, mess hall issues.
 -  **Furniture / Carpentry:** Door lock, bed, study table repair.
 -  **Misconduct / Disciplinary:** Noise disturbance, rule violations.
- **Camera Photo Evidence Upload:** Take photo directly from mobile camera or upload from gallery as proof.
- **Real-Time Status Timeline Badges:**
 -  **Pending:** Complaint raised, waiting for warden review.
 -  **In Progress:** Warden assigned maintenance staff (e.g. Electrician Ramesh working).
 -  **Resolved:** Work finished & verified.
 -  **Rejected:** Invalid or duplicate request with warden reason.
- **Direct Complaint Chat Thread:** Chat and add comments on specific complaints directly with the Warden or assigned Technician.

4 Attendance Tracker & Monthly Calendar

Student Attendance Dashboard ATTENDANCE ENGINE

- **Real-Time Today's Attendance Status:** Instant status check whether marked  **Present**,  **Absent**, or  **Outstation Leave** for the day.
- **Monthly Attendance Calendar Grid:** Interactive calendar showing daily attendance history with color-coded day badges.
- **Attendance Percentage Gauge:** Visual percentage ring (e.g., 94% Attendance) to track semester attendance requirement.

5 Mess Operations, Daily Menu & 5-Star Feedback

Daily Mess Menu & Food Quality Rating MESS MODULE

- **Daily & Weekly Mess Menu:** View today's and full week's meal schedule:
 -  **Breakfast:** 7:30 AM - 9:00 AM
 -  **Lunch:** 12:30 PM - 2:00 PM
 -  **Evening Snacks:** 5:00 PM - 6:00 PM
 -  **Dinner:** 8:00 PM - 9:30 PM
- **1-to-5 Star Meal Feedback Submission:** Submit star ratings and food quality comments for any meal.
- **Skip Meal Requests:** Mark meal skip requests before cutoff time for food wastage control.

6 Real-Time Chat & Community Messaging

⚡ Socket.io Instant Messaging System REAL-TIME CHAT

- **1-on-1 Direct Messaging:** Direct private chat channel with Hostel Warden and other Students.
- **Hostel Block & Floor Group Chat Rooms:** Group chat channels for all residents of specific hostel blocks (e.g., *Boys Block A Main Chat*).
- **Real-Time Socket.io Delivery:** Instant message delivery without page refresh.
- **Unread Message Counter Badges:** Dynamic red unread badge counters on chat tabs.
- **Media & Attachment Sharing:** Share photos and documents inside chat.

7 Announcements, Circulars & Push Alerts

🔔 Live Notification & Notice Board ALERTS

- **Official Hostel Notice Board:** View official circulars and notices published by Warden and Management.
- **Real-Time Push Notifications:** Instant alert popups for:
 - Complaint Status Updates (e.g. *"Your plumbing complaint #104 has been marked Resolved"*)
 - New Official Announcements
 - Warden Direct Messages

8 Mobile App & Desktop Web Resolution Previews

